Database Systems

Session 3

From ER Model to Relational Schema

We have an ER model. Now what?

ER → Relational Schema → SQL

Today's Question

BRIDGE

How do we turn an ER model into tables?

Entity

→ ?

Attribute

→ ?

Relationship

→ ?

Key

→ ?

M:N

→ ?

By the end: you can map a small ER model to a relational schema.

Last Week

From the real world to a conceptual model

Entity

A distinguishable real-world object

Attribute

A property of an entity

Relationship

An association among entities

ER Model

A conceptual representation

Quick recall: Give one example of each.

ER Model ≠ Relational Schema

BRIDGE

Two different levels of design

ER Model

What does the system mean?

- entities
- attributes
- relationships
- cardinalities

Relational Schema

How is the data structured as relations?

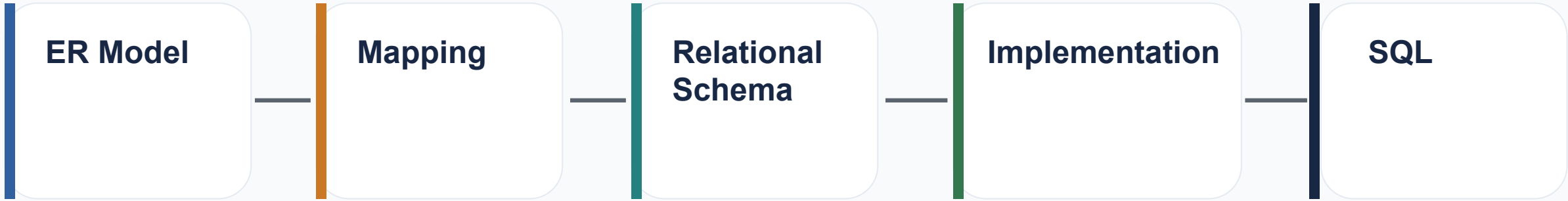
- relations
- columns
- keys
- foreign keys

Conceptual Design → Logical Design

From Meaning to Structure

BRIDGE

The bridge is mapping.

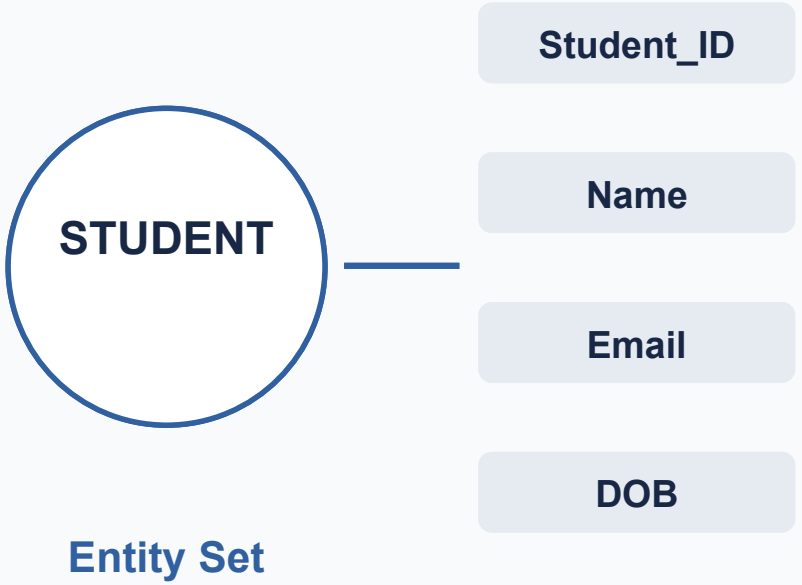


TODAY

ER → Relational Schema

Rule #1 — Entity Set → Relation

A strong entity set becomes a relation (table).



Entity Set

```
STUDENT(
  Student_ID,
  Name,
  Email,
  DOB
)
```

Relation / Table

Rule #2 — Attribute → Column

Simple attributes become columns.

ER

Student

- Student_ID
- Name
- Email
- DOB

Relational Schema

```
STUDENT(
    Student_ID,
    Name,
    Email,
    DOB
)
```

Rule #3 — Key Attribute → Primary Key

The identifier becomes the primary key.

STUDENT

Student_ID PK
Name
Email
DOB

Primary Key

- Uniquely identifies a row
- Cannot be NULL
- Identifies one entity instance

Name ≠ Identifier

Student_ID = Identifier

Composite Attributes

Decompose them into their components.

ER

Name
- First_Name
- Last_Name

```
STUDENT(
  Student_ID,
  First_Name,
  Last_Name
)
```

Rule: store the meaningful components.

Multivalued Attributes

One entity can have multiple values.

Problem

Student → Phone Numbers

How many phone columns?

✕ **Phone1 | Phone2 | Phone3**

```
STUDENT(
  Student_ID,
  Name
)

STUDENT_PHONE(
  Student_ID,
  Phone
)

PK = (Student_ID, Phone)
```

Separate relation

Quick Check

CHECK

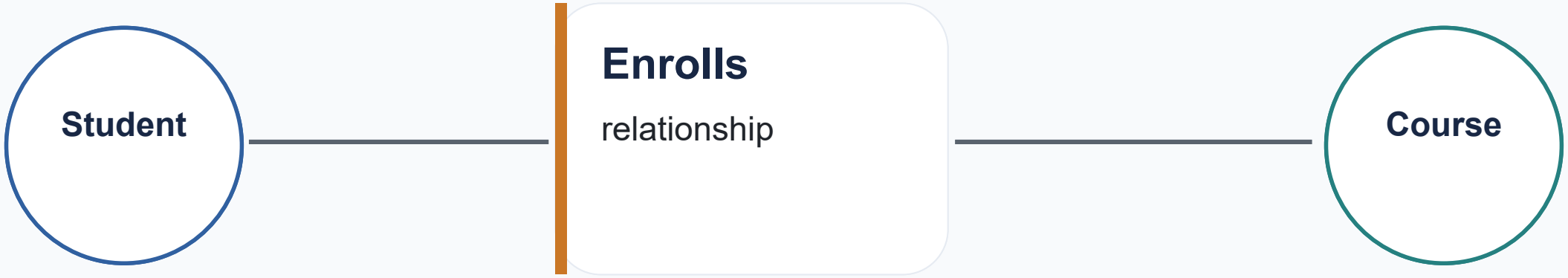
Decide: column, separate relation, or derived?

Attribute	Mapping
Student_ID	Column + PK
Name = First + Last	Multiple columns
Phone Numbers	Separate relation
Age from DOB	Derived / usually not stored

Think → pair → answer

Relationships: The Real Challenge

Entity mapping was easy. What about relationships?



Where does the relationship go?

1:N Relationship

One department offers many courses.



Question: where should the foreign key go?

1:N — Foreign Key Goes to the N-Side

The N-side stores the reference to the 1-side.

```
DEPARTMENT(
  Dept_ID PK,
  Name
)
```

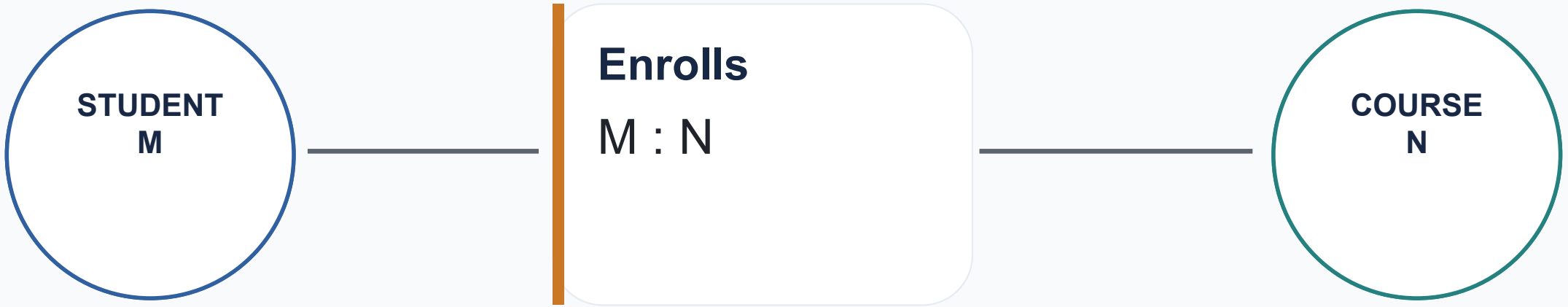
```
COURSE(
  Course_ID PK,
  Title,
  Credits,
  Dept_ID FK
)
```

FK → N-side

Why? A department can have many courses.

M:N Relationship

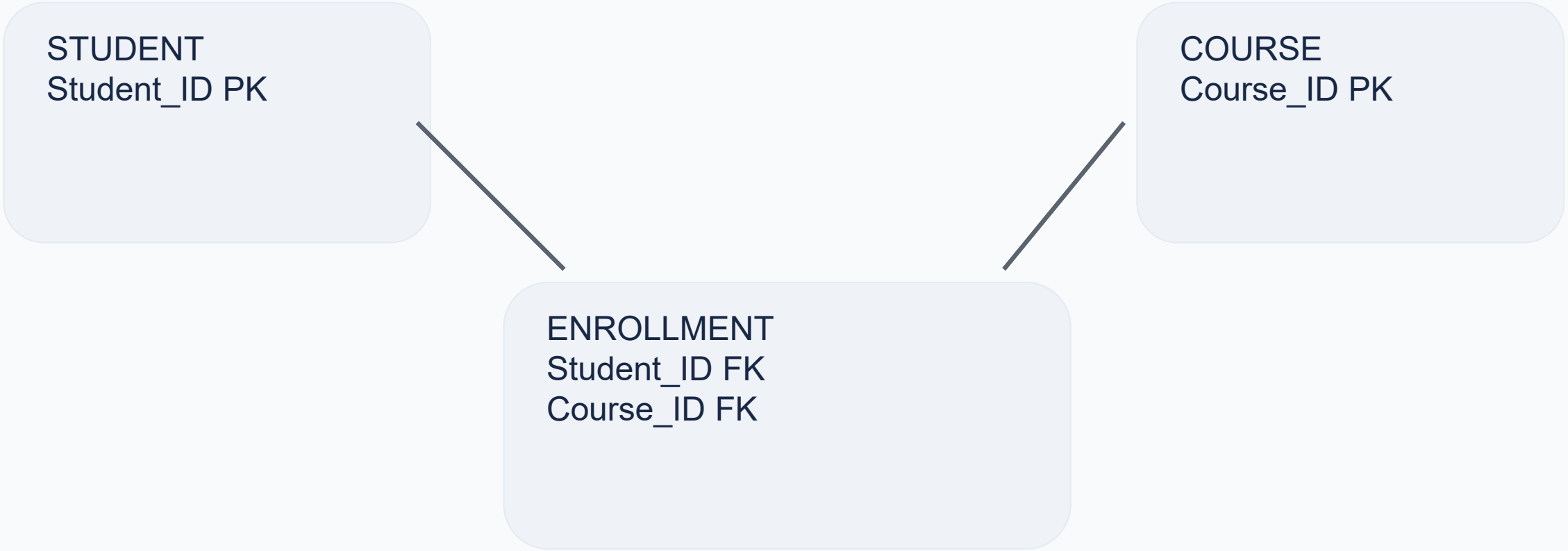
Many students can take many courses.



Can one foreign key represent an M:N relationship?

M:N → Associative Relation

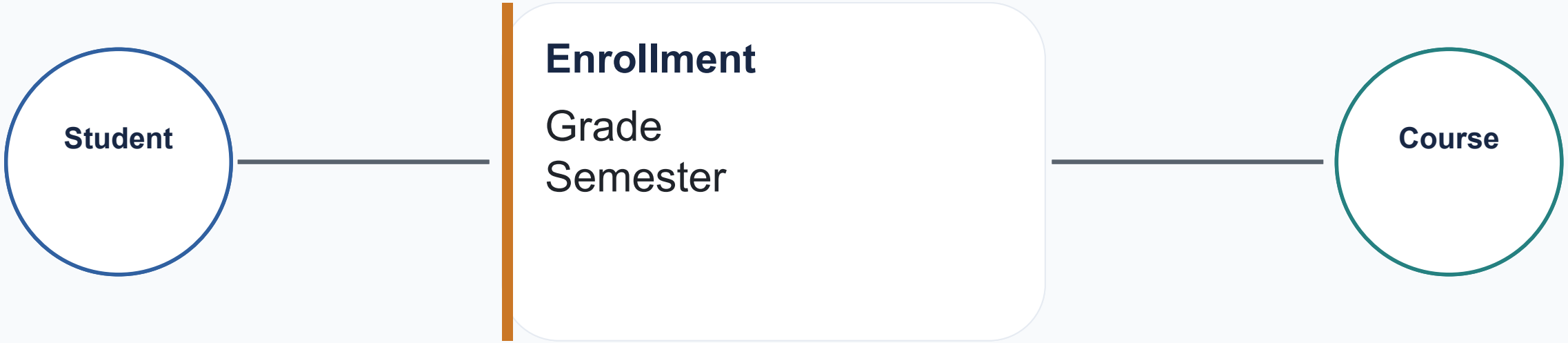
Create a new relation for the relationship.



M:N becomes two 1:N relationships.

Relationship Attributes

Grade belongs to the enrollment, not to Student or Course.




```
ENROLLMENT(
  Student_ID,
  Course_ID,
  Semester,
  Grade
)
```

Choosing the Primary Key

Ask: what uniquely identifies one enrollment?

Ali | DB | Fall 2026 | 8
Ali | DB | Fall 2027 | 19

Candidate key

(Student_ID, Course_ID, Semester)

If retaking a course is possible, Student_ID + Course_ID may not be enough.

Design depends on system rules.

1:1 Relationship

There can be more than one valid mapping.




```
EMPLOYEE(
  Employee_ID PK,
  Name,
  Parking_ID FK
)
```

FK placement depends on semantics + participation.

ER → Relational Mapping Rules

SUMMARY

A compact reference — use the rules to reason, not just memorize.

ER Concept	Relational Concept
Entity Set	Relation / Table
Attribute	Column
Key Attribute	Primary Key
Composite Attribute	Multiple Columns
Multivalued Attribute	Separate Relation
1:N	FK on N-side
M:N	New Relation
Relationship Attribute	Attribute of relationship relation

University Registration

CASE STUDY

Let's apply all the mapping rules.

Entities

Student
Course
Professor
Department

Relationships

Student → enrolls → Course

Professor → teaches → Course

Department → offers → Course

Your questions

Which are 1:N?

Which is M:N?

Where do FKs go?

Where do Grade + Semester go?

Let's Map It

Start with entities. Then add keys, foreign keys, and relationship relations.

STUDENT(...)

COURSE(...)

PROFESSOR(...)

DEPARTMENT(...)

Ask, in order:

1. What are the PKs?
2. Which relationships are 1:N?
3. Which relationship is M:N?
4. Where do the FKs go?
5. Where do Grade + Semester go?

20-Minute Design Challenge

ACTIVITY

Work in groups of 2–3.

Create the relational schema

✓ At least 4 relations
✓ Primary keys
✓ Foreign keys
✓ Enrollment relationship
✓ Grade + Semester
✓ Professor–Course relationship

Be ready to defend one decision

Why is this FK here?

Why is this a new relation?

Why is this the primary key?

What assumption did you make?

20 minutes

Compare & Challenge

ACTIVITY

One group explains. Another group challenges one design decision.

Group A

Explain your schema.

Focus on:
- PKs
- FKs
- relationship mapping
- one design decision

Group B

Find one possible problem or question.

Examples:
“Why is Semester in the PK?”
“Why is Dept_ID in COURSE?”

Engineering skill: don't only give an answer — defend the design.

One Possible Solution

One valid design under our stated assumptions.

```
STUDENT(
  Student_ID PK,
  Name,
  Email,
  DOB
)

DEPARTMENT(
  Dept_ID PK,
  Name
)
```

```
COURSE(
  Course_ID PK,
  Title,
  Credits,
  Dept_ID FK
)

PROFESSOR(
  Professor_ID PK,
  Name,
  Dept_ID FK
)
```

```
ENROLLMENT(
  Student_ID PK/FK,
  Course_ID PK/FK,
  Semester PK,
  Grade
)
```

Database English

ENGLISH

Read → Find → Explain

Key terms

relation
attribute
mapping
key
constraint
primary key
foreign key

Reading task

“Find the rule for mapping an entity set to a relation."

Then explain it in your own words.

Question:
What does “mapping” mean here?

Goal: documentation literacy — not line-by-line translation.

Presentation System

Your presentation is part of the course.

Research —— **Understand** —— **Explain** —— **Defend**

Your presentation

10–15 minutes

Problem → Why it matters → Core concept → Example/Demo → Trade-offs → Takeaways → Questions

Core principle

Don't just explain what X is.

Explain:
- why X exists
- what problem it solves
- when we should use it

Early groups receive a small incentive.

Presentation Topics

Topics are ordered around the course roadmap.

DB Engineering

- ❑ Indexing in Practice
- ❑ Transactions in Real Apps
- ❑ Database Security
- ❑ Distributed Databases

Modern DBs

- ❑ MongoDB vs PostgreSQL
- ❑ Redis
- ❑ Graph Databases
- ❑ DuckDB & Columnar Storage
- ❑ Time-Series Databases

Data Engineering

- ❑ Apache Spark Kafka
- ❑ Warehouse / Lake / Lakehouse
- ❑ dbt & the Modern Data Stack

Database + AI

- ❑ ORMs & Schema Migrations
- ❑ Vector Databases & RAG
- ❑ Text-to-SQL & SQL Agents

Topics answer engineering questions — not just “What is X?”

Presentation Workflow

Each group produces reusable learning material.



Coordinator collects

Slides • resources • 2 questions + answers • optional code/demo

Question Bank & Assessment

Presentation work feeds the shared study resources.

Question Bank

Each group submits:
- 2 questions
- answers
- sources/resources

The coordinator shares the materials with the class.

Presentation — 2 points

Technical Content	0.60
Understanding	0.35
Practical Example/Demo	0.30
Presentation Skills	0.30
Teamwork	0.20
Q&A	0.15

Some final-exam questions will be conceptually based on the shared question bank.

What Can We Do Now?

A checkpoint before we move to SQL.

- Map entities to relations
- Map attributes to columns
- Choose primary keys
- Place foreign keys
- Map 1:N and M:N relationships
- Handle relationship attributes
- Explain design decisions

Next step

Relational Schema

↓

SQL

↓

Implementation

Traditional foundation → practical implementation

Homework

Practice the mapping process — no SQL yet.

Choose one

Library
Hospital
Restaurant
Online Shop
Gym
Taxi

ER → Relational Schema

Minimum:

- 4 entities
- 3 relationships
- Primary keys
- Foreign keys
- At least 1 design decision

Explain one decision

Example:

Why is the FK here?

Why is this relationship a separate relation?

Why is this the PK?

No SQL yet.